Choosing a Fast Way to Add
Answer Key

Kindergarten–Grade 1

Quick Answers

1. 9	4. 9	7. 4	10. 10
2. 8	5. 9	8. 8	
3. 10	6. 3	9. 8	

Common wrong answers

9. *If they got 7: said the larger part itself as the first count instead of the next number, so the total is one too small*
10. *If they got 5: added only the first two parts and forgot the third*

1. $7 + 2$ — best way: count on.

Step 1. One part (2) is tiny and the other (7) is large, so counting on beats counting everything: hold 7.
Step 2. Say the *next* two numbers: "8, 9." Two counts, so $7 + 2 = \boxed{9}$.
Accept also: a student who makes ten first ($7 + 2$ is one less than $7 + 3$) still gets 9; the count-on is the shortest route.

2. $4 + 4$ — best way: double.

Step 1. Both parts are the same number, so this is a known double — no counting needed.
Step 2. Double 4 is $\boxed{8}$.

3. $6 + 4$ — best way: make ten.

Step 1. 6 and 4 are ten-partners: on a ten-frame, 6 counters leave exactly 4 empty spaces, so the frame fills up.
Step 2. A full frame is ten, so $6 + 4 = \boxed{10}$.

4. $3 + 6$ — best way: count on from the larger part.

Step 1. Order does not change a sum, so start from the *larger* part, 6, and count on the smaller part, 3. (Starting at 3 would take six counts instead of three.)
Step 2. Say "7, 8, 9" — three counts — so $3 + 6 = \boxed{9}$.

5. $4 + 5$ — best way: near double.

Step 1. The parts are neighbours, so lean on the double you already know: $4 + 4 = 8$.
Step 2. 5 is one more than 4, so the sum is one more than 8: $4 + 5 = \boxed{9}$.

6. $7 + \_\_\_\_ = 10$ — best way: ten-partners.

Step 1. The whole is 10 and one part is 7, so the missing part is what is left of the ten-frame after 7 counters.
Step 2. Count the empty spaces: "8, 9, 10" — three of them. The missing part is $\boxed{3}$.

7. A number added to itself gives 8 — best way: doubles.

Step 1. "Added to itself" means both parts are the same, so ask which double makes 8.
Step 2. Run the doubles: $3+3=6$, $4+4=8$. Kim added $\boxed{4}$.

8. ____ $+2=10$ — best way: ten-partners.

Step 1. The whole is 10 and the known part is 2, so the missing part is the ten-partner of 2.
Step 2. Ten-partners come in pairs, and 2 pairs with 8 because $8+2=10$. The missing part is $\boxed{8}$.

9. Sam's error on $2+6$.

Step 1. Sam picked the right strategy: hold the larger part, 6, and count on the smaller part, 2.
Step 2. His mistake is *where he started counting*. He said the 6 itself as his first count, so he only moved one step forward and landed on 7. The first thing you say must be the *next* number after the one you are holding.
Step 3. Counting correctly: "7, 8" — two counts, one for each of the 2. So $2+6=\boxed{8}$.

10. Challenge: $2+3+5$.

Step 1. Three parts may be added in any order, so scan for a pair that makes a friendly number before adding left to right.
Step 2. 2 and 3 make 5, and 5 is the ten-partner of the remaining 5 — so group them as $(2+3)+5$.
Step 3. $(2+3)+5=5+5$, a double that is also a ten-partner pair, so the total is $\boxed{10}$.
Also correct: $2+(3+5)=2+8$, counted on, gives the same 10; credit any order with correct work.